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Abstract

die Zusammenfassung

Preface

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1 Introduction

1.1 Motivation

An industrial robot arm can run the same pick-and-place motion in two scenes that look nearly identical, yet only one is hazardous, for example a vial with its cork missing and a corrosive pictogram, or a knife offered blade-first instead of handle-first. Generic object detection misses these small, local cues, so this thesis builds a scene graph of the work cell from the robot's cameras and an agentic framework that annotates it with evidence-backed risk information: object states, label content, object parts, and the relations that turn them into hazards.

1.2 Task and Requirements

This thesis addresses the following tasks and requirements:

1.2.1 Literature Review

Review scene graph generation, open-vocabulary perception, LLM- and VLM-based agents, and risk assessment in manipulation, building on the institute's existing risk-annotated scene graph framework.

1.2.2 Schema Design

Design a scene graph schema for the pick-and-place work cell (robot, object parts, object states, labels) and a separate schema for risk annotations.

1.2.3 Scene Graph Generation

Implement a pipeline that builds and updates the scene graph from the robot's two RGB-D cameras (wrist and third-person) while the pick-and-place task runs, live or from recorded ROS bags.

1.2.4 Agentic Risk Annotation

Design and implement an agentic framework that annotates the graph with risk-relevant facts and hazards, supported by evidence and confidence, and exports them as risk triples.

1.2.5 Evaluation

Evaluate the framework against baselines on three controlled pick-and-place scenarios, including the chemical vial and cutlery scenarios.

1.2.6 Documentation

Document the pipeline, schemas, data, and results so that another researcher can reproduce every result.

1.3 Contributions

1.4 Outline of the Report

2 Literature Review

3 Conception

3.1 Overview

The concept has two connected parts. The first part generates a scene graph of the pick-and-place work cell from the robot's cameras and the robot state. The second part is an agentic framework that inspects this graph and the camera images with tools and annotates the graph with risk-relevant information: object states such as a missing cork, label content such as a corrosive pictogram, object parts such as the blade of a knife, and the relations that turn these details into hazards. The annotated graph is produced while the task runs, but it does not control the robot.

The pipeline consists of five modules connected through ROS, as shown in Figure 1. A sensor layer provides time-aligned camera images, depth data and robot joint states, together with the camera poses from calibration. A perception module detects objects and object parts and estimates their 3D poses. A scene graph generator builds and updates the graph from these results. An annotation agent inspects the graph and the images and writes risk annotations. Finally, a validation and export module checks every annotation against the schema and publishes the annotated graph. Because all modules read their input from ROS topics, the pipeline runs the same way on the live system and on recorded ROS bags. Code, schemas, prompts and model settings are kept under version control, so every result can be reproduced.

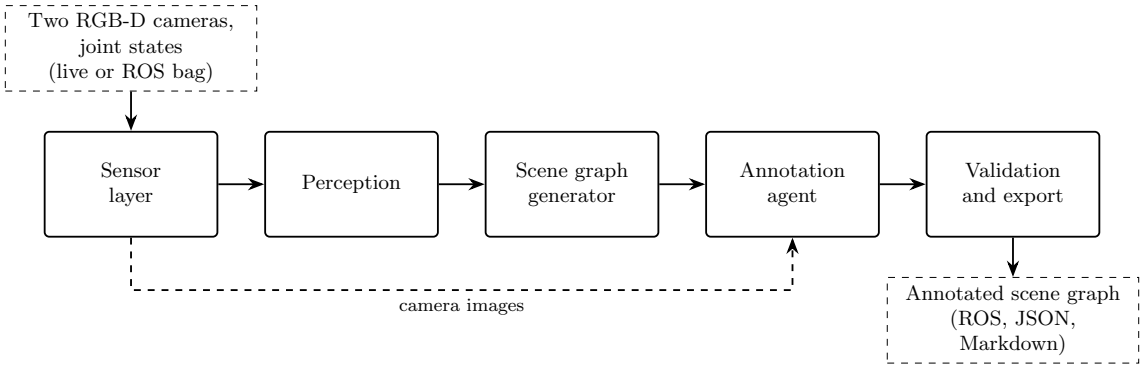


Figure 1: Architecture of the pipeline. Solid arrows show the main data flow; the dashed arrow shows that the annotation agent also inspects the camera images.

3.2 System Setup

The work cell is a tabletop with a robot arm (a Franka Emika arm or another available industrial arm), its gripper and RGB-D cameras, integrated through ROS. Two Intel RealSense D455 cameras are used: one on the wrist for close-up views and one fixed in front of the workspace for a third-person view. Because the base of the arm is fixed, the camera poses follow from the robot's kinematics and calibration, so no SLAM or visual odometry is needed.

The concept uses existing perception models and, where suitable, the institute's existing risk-annotated scene graph framework. The agentic framework is built on existing LLMs and VLMs. Training models from scratch, new motion planning or control methods, and real hazardous scenarios are out of scope.

3.3 Scene Graph Generation

The scene graph describes the work cell as objects with attributes (nodes) and relations between them (edges), for example *vial in rack* or *knife held by gripper*. A versioned schema defines the node and edge types, spatial and contact relations, persistent IDs, 3D poses, timestamps, sources and confidence values. Nodes cover the robot, objects, object parts, object states and labels.

Existing models detect and segment the objects and relevant object parts, and their 3D poses are estimated from the depth data and camera calibration. Spatial relations are computed from the 3D poses with defined rules and thresholds. The graph is updated at a fixed rate, object identities are kept across frames and occlusions, and each graph version is linked to the current phase of the pick-and-place task. The graph is published through ROS and stored as JSON, and it can be regenerated from recorded ROS bags.

3.4 Risk Annotation

Risk information is kept in a separate annotation schema that refers to the scene graph nodes. It supports unknown and novel values. A hazard is expressed as a triple, for example

```
(knife_03.blade, pointing_toward, human_hand_zone)
```

with hazard type, severity, evidence and confidence attached, and the triples are exported as BRAAIS risk triples.

The agentic framework first identifies the nodes and relations that may be relevant for risk. It then uses vision-language models and supporting tools to inspect objects, analyse spatial relationships and retrieve relevant knowledge, and it creates or updates the annotations on the graph. All tool calls and intermediate results are logged.

For each annotation, the framework records the supporting evidence, timestamps, model details, prompts and confidence. Annotations without valid evidence are rejected by schema validation, and uncertain or contradictory observations are marked as unknown. The annotated graph is published through ROS and exported as JSON or Markdown.

3.5 Evaluation Concept

The framework is evaluated on three controlled pick-and-place scenarios, with the chemical vial and cutlery scenarios reused from the BRAAIS specification. It is compared against three baselines: an unannotated scene graph, a single VLM call without tools, and a video-based VLM approach without a scene graph. The evaluation checks whether the scene graph is correct, whether the risk-relevant facts are correct, whether the annotations are supported by evidence, and whether the system is fast and reliable enough to run during the task.

For each recorded scene, reference graphs and reference annotations are created following an annotation manual that defines how objects, parts, relations and risk facts are labelled. The references are reviewed and validated against ground truth, and any discrepancies are documented. At least two LLM or VLM back-ends are tested for the agent, at least one of them hosted locally, with repeated runs, and the effect of the main design choices, such as the use of tools, abstention and the number of camera views, is analysed.

4 Implementation

5 Evaluation

6 Discussion and Limitation

7 Conclusion and Future Work

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